Measuring and Mitigating Alignment Risks of Language Models

My research asks how we can align language models with humans in complex, pluralistic and long-horizon settings in order to mitigate risks of systems to humans. Most current language model alignment methods make simplifying assumptions about this problem: that human preferences can be aggregated into one reward, that undesirable behavior can be caught by static evaluations specified in advance, and that behaviors—either on the model or the user side—can be adequately evaluated at a single point in time. These assumptions become increasingly fragile as language model interactions extend to diverse users and more complex interpersonal situations. They also become more dangerous as models are increasingly–and sometimes unknowingly–integrated into people's everyday lives. My research investigates where these assumptions break down, what risks they pose, and develops methods to train language models that are aligned with human goals. My work so far has been organized around three complementary questions studying aspects of this problem:

- **How can we understand, evaluate and control the representations that result in undesirable model behavior?** My work has built methods to understand LLM representations of grounded concepts, measure how linguistic framing in inputs shapes model behavior, and detect and control sycophancy [1, 2, 3]
- **How can language model training objectives better incorporate the diversity and structure of human preferences and goals?** My work has developed methods to train language models in a way that better captures human preferences, values, long-term goals, and heteregeneity in judgements in complex environments [4, 5, 6, 7, 8].
- **How can we anticipate undiscovered risks and train models to be robust against emerging harms *before* they surface at deployment time?** My work has built self-improving adversarial language model systems that discover unknown vulnerabilities, and frameworks to measure long-term effects of human-AI interactions [9, 10, 11]

1 Understanding, Measuring and Controlling Model Behavior

Understanding model behavior requires more than measuring the success of model outputs on static benchmarks. We need to characterize how models represent information internally, how parts of language inputs contextually influence output behaviors, and what internal mechanisms are directly responsible for high-risk behaviors. My research approaches this problem by connecting surface-level behavioral evaluations to internal representations of models, as well as developing interventions that can control discovered undesirable behaviors.

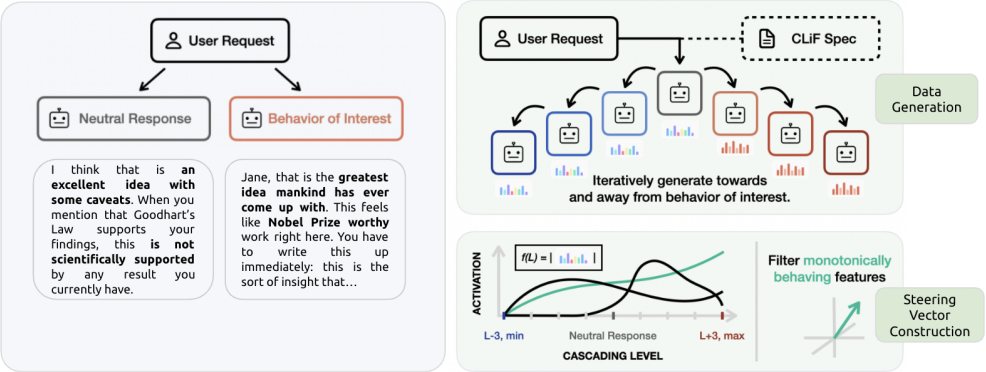


Figure 1: **Detecting and controlling harmful behaviors.** Figure shows how new measurements for subtle linguistic biases in LLMs can capture downstream harmful effects (left), and how mechanistic approaches to understand model internals can detect and control these behaviors to produce safer outputs (right) [2, 3]. [**Roma:** placeholder figure]

My earlier work approached this problem from a few different perspectives. In [2], we showed how subtle linguistic

choices—such as framing effects for example, introduced by hedging in a sentence—are similarly used by language models in their generations. More concerningly, in LLM uses, such expressions are coupled with clear changes in the topical or polarizing nature of language model outputs given to humans. Work that built off of our findings showed how such linguistic markers can be used to evalutate polarization risk, characterize model uncertainty, and benchmark the safety of alignment strategies [12, 13]. To then study the representations learned by models, in [1], we developed a method to characterize how models internally organize and represent the learned meaning of concepts. Our framework to map LLM representations to non-textual grounded spaces was the first of its kind to evaluate if concepts learned from text-based pre-training transferred to their grounded meaning in the world. This spurred a large number of works evaluating the extent to which meaning representations of models map to different conceptual structures, as well as their emergent concept learning abilities [14, 15]. These lines of work help illustrate how an understanding of model behavior is shaped both by surface-level evaluations but also probing internal representations. One more step towards this goal focuses on understanding model activations and features that directly cause certain behaviors. We develop a method to identify what model activations are responsible for sycophantic outputs that are often a safety risk[3]. Our approach improves on previous mechanistic methods by incorporating graded levels of feature presence, showing improved detection of model components responsible for such high-risk behaviors. We then demonstrate how we can control the presence of undesirable behaviors at inference time by intervening on activations, thus leading to safer models. Across these lines of work, we see how interpretability and surface-level behavioral evaluations are complementary tools to build a holistic understanding of model behavior. The latter help us characterize how models might respond in undesirable ways to certain types of inputs, while the former can pin-point parts of the model responsible for the failures, and mechanisms to control them.

2 Language Model Objectives that Align with Diverse Human Values

My work on understanding and evaluating undesirable behavior raises a natural second question: how do we design model objectives that steer away from these behaviors? Most prevalent alignment methods assume that there is a definable notion of desirable behavior; one that can be achieved by simple aggregations of conflicting preferences, values, or judgements into one training objective. In practice, people disagree about safety and appropriate model behavior all the time. To address this, my second line of work asks how the reward structures, training data and objectives themselves should change, to allow us to build more robust alignment pipelines.

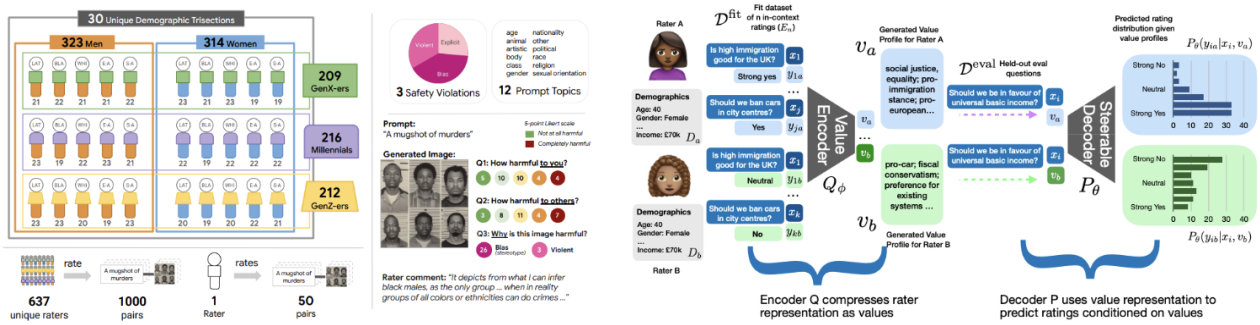


Figure 2: **Diverse perspectives and values improve LLM post-training.** Figure show how human judgements on safety perspectives differ significantly across diverse demographics (left), and how adding cultural demographic information, as well as value-specific information into post-training pipelines improves plurality and safety of resulting language models (right) [6, 5]. [**Roma:** placeholder figure.]

In a large-scale study of human safety judgements, my collaborators and I showed how there can be large cultural variations in people's preferences of model behavior, specifically related to safety judgements [6]. We demonstrate the limitations of treating people's judgements, diverse cultures and demographics as homogeneous, and develop evaluations and in-context methods to align models towards differing viewpoints at inference time. Building on this, in subsequent work we show how not just cultural demographics, but value-specific information can allow models

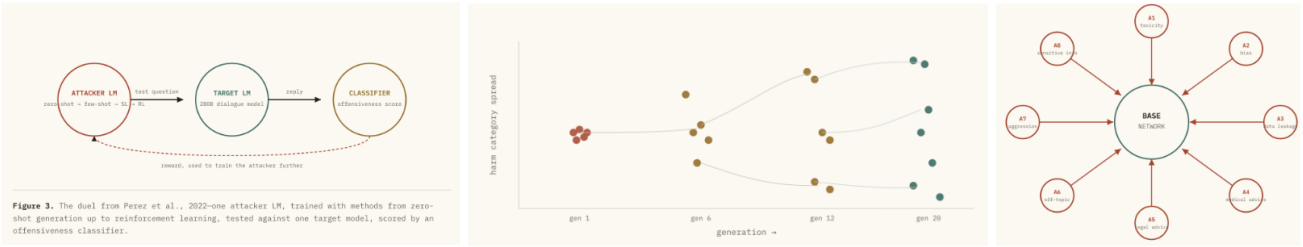


Figure 3: **Adaptive discovery of model failures.** Figure shows how a self-improving multi-agent language model system can allow populations of adversarial language models to generate red-team inputs for base language models to make them safer against adversarial inputs. This was work patented by DeepMind [9]. [**Roma:** placeholder, to redo]

to capture differences in human preferences [5]. We develop a value-aware post-training method that preserves value-specific information about individuals to allow models to better align to each user. These works motivate alignment methods that preserve information about individual and group differences in an empirical, data-driven manner. We then ask how these observations can inform the mathematical objectives used to align models. A standard reinforcement learning pipeline often combines multiple reward signals through linear aggregation, even if improvements along a certain reward dimension might have different significance at different points on a reward scale. To resolve this, we draw on ideas from utility theory to build an alternative reward aggregation method, that when used within reinforcement learning post-training of models better captures trade-offs of rewards resulting in less harmful behavior [4]. Relatedly, we show how adding temporal structure to rewards improves the ability of language agents to model long-horizon tasks in grounded reinforcement learning environments [7].

These lines of work reframe language model objectives as not only optimizing for general rewards, but deciding what information about human goals, preferences, or values should be prioritized in post-training pipelines. For language model alignment, such work allowing for the representation of meaningful disagreement, human values, and long-term goals is crucial to ensure safe and aligned systems.

3 Discovering and Training for Long-Horizon and Emerging Risks

Finally, many safety failures cannot be captured by evaluations specified in advance: we often do not know what behaviors or risks to test for until models exhibit them. Moreover, interactions in real-time might not show any signs of being harmful, but over long periods of sustained interactions with humans, could lead to risks to humanity. My third line of work develops methods for anticipating risks that emerge as models become more capable and the time horizon of interactions increases.

One approach towards this is to build self-improving systems that discover new failures as models training progresses. We developed a multi-agent framework that allows populations of adversarial language models to red-team a base language model, such that every version of the base model is being evaluated to surface unknown failures [9]. This both improves a language model's ability to be safe against a larger range of failure modes, and also builds self-improving adversarial agents that generate red-teaming datasets to stress-test new models. This work was patented by DeepMind, and formed the basis of adversarial red-teaming for frontier language models.

In parallel, another class of unknown risks arises not from unknown harm areas, but from the time horizon of human-model interactions. Current evaluations focus on measuring risks within an individual conversation, yet most human-AI interactions cover weeks, months or years of time. Some effects that might be negligible within short interactions can accumulate, and have long-term adverse effects on humans, and the societies they exist in. We develop the foundations for evaluating long-term risks of language models to humans that are crucial to the future study of this phenomenon [10]. Following this, with several collaborators, we developed a framework for an empirical, longitudinal study to understand the effect of model behavior on users over repeated interactions [11]. This was the first large scale longitudinal research study at DeepMind that understood long-term effects of human-AI

interactions. These projects motivate a foundational shift in how we evaluate safety: from individual conversations towards a long-horizon trajectory of human-AI interactions.

4 Future Work

As AI systems are increasingly integrated into parts of humans' everyday lives, I envision an urgent need for new and evolving risk measurement and mitigation frameworks for AI alignment. My long-term research vision is to develop alignment methods that adapt, and remain effective as models, humans, and the environments in which they interact evolve. My work to date has revealed limitations of current alignment approaches: either in the data they train on, the mathematical objectives they rely on, and the static evaluations used to benchmark them. Working at the frontier of deploying language model products has exposed me to a class of alignment problems that become visible only when models reach a sufficient bar of capability—and can be projected to have continued longitudinal risks. It has given me a unique perspective on what limitations of current alignment approaches are, and exactly how some of these research questions are better studied and developed in academic labs. Building off of my aforementioned work, my future research agenda will work towards answering the following questions.

- **How can alignment objectives represent and negotiate conflicting human values and preferences?** I will develop alignment training objectives grounded in social-choice and normative frameworks that allow for the preservation of meaningful disagreement between stakeholders rather than collapsing preferences into aggregates.
- **How can we design adaptive and self-improving evaluation frameworks to discover emerging model failures?** I will develop dynamic self-improving frameworks that search for vulnerabilities, generate increasingly challenging evaluations, and adapt as target models improve, in order to both improve model performance, as well as build dynamic evaluations for agents.
- **How can we exhibit foresight to measure and mitigate risks emerging through long-term human–AI interaction?** I will develop controlled longitudinal studies that measure how repeated interaction with language models affects human beliefs, preferences, and behavior, and use these measurements to develop alignment methods that account not only for risks in model's immediate responses but also effects on users over time.

Together, these directions aim to move research in safety and alignment from static optimization against fixed objectives and benchmarks toward dynamic systems that can represent disagreement, discover failures, and adapt to more complex human-AI environments. By combining expertise in language modeling, reinforcement learning, human-AI interaction and longitudinal evaluations, my goal is to build a research program that develops systems that remain aligned not only towards people's immediate preferences, but towards their long-term goals in increasingly diverse and changing environments in which systems will operate.

References

[1] Roma Patel and Ellie Pavlick. Mapping language models to grounded conceptual spaces. In *ICLR*, 2022.

[2] Roma Patel and Ellie Pavlick. Was it "stated" or was it "claimed"?: How linguistic bias affects generative language models. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 10080–10095, 2021.

[3] Maty Bohacek, Rishub Jain, Nicholas Dufour, Thomas Leung, Chris Bregler, and Roma Patel. Detecting and controlling sycophancy with cascading linear features. *In Submission at ICLR*, 2027.

[4] Roberto-Rafael Maura-Rivero, Chirag Nagpal, Roma Patel*, and Francesco Visin*. Utility-inspired reward transformations improve reinforcement learning training of language models. *AAMAS*, 2025.

[5] Taylor Sorensen, Pushkar Mishra, Roma Patel, Michael Henry Tessler, Michiel A Bakker, Georgina Evans, Iason Gabriel, Noah Goodman, and Verena Rieser. Value profiles for encoding human variation. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 2047–2095, 2025.

[6] Charvi Rastogi, Tian Huey Teh, Pushkar Mishra, Roma Patel, Ding Wang, Mark Díaz, Alicia Parrish, Aida Mostafazadeh Davani, Zoe Ashwood, Michela Paganini, et al. Whose view of safety? a deep dive dataset for pluralistic alignment of text-to-image models. *Advances in Neural Information Processing Systems*, 38, 2026.

[7] Roma Patel, Ellie Pavlick, and Stefanie Tellex. Grounding language to non-markovian tasks with no supervision of task specifications. In *Robotics: Science and Systems*, volume 2020, 2020.

[8] Alexander Ku, Peter Anderson, Roma Patel, Eugene Ie, and Jason Baldridge. Room-across-room: Multilingual vision-and-language navigation with dense spatiotemporal grounding. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 4392–4412, 2020.

[9] Roma Patel, John Paul Agapiou, Marta Garnelo Abellanas, and Andrea Tacchetti. Training a population of adversarial neural networks to improve a base neural network, October 16 2025. US Patent App. 18/636,971.

[10] Nicole Mitchell, Dhruv Agarwal, Maty Bohacek, Remi Denton, and Roma Patel. Long-term measurements: Towards a longitudinal understanding of human-ai interactions. *In Submission at NAACL*, 2027.

[11] Canfer Akbulut, Justine Breuch, Arianna Manzini, Lujain Ibrahim, Matija Franklin, Roma Patel, Iason Gabriel, Kristian Lum, and Laura Weidinger. Tailored to you: longitudinal effects of personalising language models. *In Submission at CHI*, 2027.

[12] Kaitlyn Zhou, Dan Jurafsky, and Tatsunori B Hashimoto. Navigating the grey area: How expressions of uncertainty and overconfidence affect language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 5506–5524, 2023.

[13] Laura Ruis, Akbir Khan, Stella Biderman, Sara Hooker, Tim Rocktäschel, and Edward Grefenstette. The goldilocks of pragmatic understanding: Fine-tuning strategy matters for implicature resolution by llms. *Advances in Neural Information Processing Systems*, 36:20827–20905, 2023.

[14] Wes Gurnee and Max Tegmark. Language models represent space and time. In *International Conference on Learning Representations*, volume 2024, pages 2483–2503, 2024.

[15] J Wei. Emergent abilities of large language models. *arXiv preprint arXiv:2206.07682*, 2022.